

Object Oriented Programming in C++

BCA/BIT/BE-Computer/PGDCA
2nd Semester

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History Of C++

- The C programming language was devised in the early 1970s by Dennis M. Ritchie an employee from AT&T Bell Labs. (*American Telephone and Telegraph*)
- Many other programming languages are derived from the C language. The languages Objective-C and C++ for instance are derived from the C language. Both languages add the “object oriented” element to the language C.
- The programming language C++ (originally named “C with Classes”) was devised by Bjarne Stroustrup also an employee from Bell Labs (AT&T). Stroustrup started working on C with Classes in 1979.

History of C++ cont...

- He chose to enhance the C language with Simula-like features, because the C language was fast and portable. Stroustrup did not only use features from Simula but also borrowed features from the languages Ada, CLU, ALGOL 68 and ML.
- In 1983, the name of the language was changed from C with Classes to C++. The ++ is C language operator and C++ is the incremented version of C. The first commercial release of the C++ language was in 1985.
- In 1998, a joint ANSI-ISO committee standardized the C++ language.

Object Oriented Programming (OOP) Vs Process Oriented Programming (POP)

OOP	POP
In object oriented programming, program is divided into small parts called objects.	In procedural programming, program is divided into small parts called functions.
Object oriented programming follows bottom up approach	Procedural programming follows top down approach
Object oriented programming have access specifiers like private, public, protected etc.	There is no access specifier in procedural programming
Adding new data and function is easy	Adding new data and function is not easy
Object oriented programming provides data hiding so it is more secure	Procedural programming does not have any proper way for hiding data so it is less secure
Overloading is possible in object oriented programming	In procedural programming, overloading is not possible
In object oriented programming, data is more important than function	In procedural programming, function is more important than data
Object oriented programming is based on real world.	Procedural programming is based on unreal world
Examples: C++, Java, Python, C# etc.	Examples: C, FORTRAN, Pascal, Basic etc.

Concept of OOP

- Object-oriented programming aims to implement real-world entities like inheritance, hiding, polymorphism, etc in programming. The main aim of OOP is to bind together the data and the functions that operate on them so that no other part of the code can access this data except that function.

Characteristics:

- Class
- Objects
- Encapsulation
- Abstraction
- Polymorphism
- Inheritance
- Dynamic Binding
- Message Passing

Class

- The building block of C++ that leads to Object-Oriented programming is a Class. It is a user-defined data type, which holds its own data members and member functions, which can be accessed and used by creating an instance of that class. A class is like a blueprint for an object.
- Example: Car is the class and wheels, speed limits, mileage are their properties.

Object

- An Object is an identifiable entity with some characteristics and behaviour. An Object is an instance of a Class. When a class is defined, no memory is allocated but when it is instantiated (i.e. an object is created) memory is allocated.
- Example: **Fruit** Mngo; where **Fruit** is a class and Mango is an Object.

Encapsulation

- In normal terms, Encapsulation is defined as wrapping up of data and information under a single unit. In Object-Oriented Programming, Encapsulation is defined as binding together the data and the functions that manipulate them.

- Example:

```
class person
{
    private:
    char name[20];
    int id;
    public:
    void getdetails() {}
};
```

Abstraction

- Data abstraction is one of the most essential and important features of object-oriented programming in C++. Abstraction means displaying only essential information and hiding the details. Data abstraction refers to providing only essential information about the data to the outside world, hiding the background details or implementation.

Polymorphism

- The word polymorphism means having many forms. In simple words, we can define polymorphism as the ability of a message to be displayed in more than one form.
- A person at the same time can have different characteristic. Like a man at the same time is a father, a husband, an employee. So the same person possesses different behavior in different situations. This is called polymorphism.
- Example: Function Overloading and Operator Overloading

Inheritance

- The capability of a class to derive properties and characteristics from another class is called Inheritance. Inheritance is one of the most important features of Object-Oriented Programming.
- **Sub Class:** The class that inherits properties from another class is called Sub class or Derived Class.
- **Super Class:** The class whose properties are inherited by sub class is called Base Class or Super class.
- **Reusability:** Inheritance supports the concept of “reusability”, i.e. when we want to create a new class and there is already a class that includes some of the code that we want, we can derive our new class from the existing class. By doing this, we are reusing the fields and methods of the existing class.
- Example: Dog, Cat, Cow can be Derived Class of Animal Base Class.

Dynamic Binding

- In dynamic binding, the code to be executed in response to function call is decided at runtime.
- Example: C++ has virtual functions to support this feature.

Message Passing

- Objects communicate with one another by sending and receiving information to each other. A message for an object is a request for execution of a procedure and therefore will invoke a function in the receiving object that generates the desired results. Message passing involves specifying the name of the object, the name of the function and the information to be sent.
- Example:

```
class ABC
{
    void Method1(ABC obj)
    {
        // Method does something which
        //you assigned to do
    }
};
```

```
int main()
{
    ABC objA,objB
    objA.Method1(objB);
}
```

